CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ

Chapter 4: Linear Inequations (In One Variable)

Previous year board questions, topic-wise: 9 questions

The tag after each question gives the board examination year and the marks. Solutions are in a separate PDF with the same question numbers.

1 Solution set and the replacement set

1. Find the solution set of $0 < -\frac{x}{3} < 2, x \in Z$. (a) $\{-5, -4, -3, -2, -1\}$ (b) $\{-6, -5, -4, -3, -2, -1\}$ (c) $\{-5, -4, -3, -2, -1, 0\}$ (d) $\{-6, -5, -4, -3, -2, -1, 0\}$ ICSE 2025, 1 mark

2. Find the solution set of $2x + 4 \leq 14$, where x is a whole number. (a) $\{1, 2, 3, 4, 5\}$ (b) $\{0, 1, 2, 3, 4, 5\}$ (c) $\{1, 2, 3, 4\}$ (d) $\{0, 1, 2, 3, 4\}$ ICSE 2023, 1 mark

2 Combining inequations

3. Solve $2x - \frac{5}{3} < \frac{3x}{5} + 10 \leq \frac{4x}{5} + 11, x \in R$. Write the solution set and show it on the number line. ICSE 2025, 3 marks

4. Solve $-3 + x \leq \frac{7x}{2} + 2 < 8 + 2x$, where x is an integer. Write the solution set and show it on the number line. ICSE 2024, 3 marks

5. Solve $-5(x - 9) \geq 17 - 9x > x + 2, x \in R$. Write the solution set and show it on the number line. ICSE 2023, 3 marks

6. Solve $\frac{3x}{5} + 2 < x + 4 \leq \frac{x}{2} + 5, x \in R$, and show the solution set on the number line. ICSE 2020, 3 marks

7. Solve $11x - 4 < 15x + 4 \leq 13x + 14$, where x is a whole number. Write the solution set and show it on the number line. ICSE 2019, 3 marks

8. Solve $-2 + 10x \leq 13x + 10 < 24 + 10x$, where x is an integer. Write the solution set and show it on the number line. ICSE 2018, 3 marks

9. Solve $-8\frac{1}{2} < -\frac{1}{2} - 4x \leq 7\frac{1}{2}$, where x is an integer, and show the solution set on the number line. ICSE 2017, 3 marks

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